

## DEVICE ANATOMY 2.0

Engineering Investigation of an Electronic Device at End-of-Life

|          |                                    |       |         |
|----------|------------------------------------|-------|---------|
| Course   | E-Waste & Environmental Management | Class | TE INFT |
| Duration | 45-60 minutes                      | Marks | 10      |

10  
10

### Group Details

|            |                                     |                      |                                 |
|------------|-------------------------------------|----------------------|---------------------------------|
| Group No.: | 11                                  | Device Investigated: | DVD player                      |
| Date:      | 02 Sept 2026                        | Condition:           | Non-functional                  |
| Member 1:  | Vishal R. Prajapati<br>(24101B0047) | Member 2:            | Meet M Alshi<br>(24101B0048)    |
| Member 3:  | Altamash Ansari<br>(24101B0048)     | Member 4:            | Pranitha Shetty<br>(24101B0027) |
| Member 5:  |                                     |                      |                                 |

Condition: ☐ Functional ☒ Non-functional ☐ Damaged ☐ Unknown

### Activity Evidence

During the activity, photograph all group members actively participating in the Device Anatomy investigation. Print the photograph and attach it below. A posed group selfie is not considered activity evidence.

ATTACH PRINTED GROUP PHOTOGRAPH HERE

BACK SIDE

Photo caption: Our group investigating the anatomy and end-of-life characteristics of a DVD player.

Group members visible: Vishal, Meet, Pranitha, Altamash,

#### ACTIVITY CHALLENGE

You are an e-waste engineer. Your task is to investigate the material anatomy of an electronic product, identify recovery opportunities and environmental risks, and propose engineering changes that improve its end-of-life management.

Core question: Can you make this device more circular?

## Part A — Device Profile

|                                      |                                                                                                                                                                                                                                         |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device name                          | DVD Player                                                                                                                                                                                                                              |
| Primary function                     | To play DVDs, video CDs. to display video on it's built-in LCD screen.                                                                                                                                                                  |
| Why does this device become e-waste? | <input type="checkbox"/> Obsolescence <input type="checkbox"/> Damage <input type="checkbox"/> Upgrade <input checked="" type="checkbox"/> Repair not economical <input type="checkbox"/> Failure <input type="checkbox"/> Other: _____ |

## Part B — Engineering Anatomy

Identify at least 6 major components. Do not merely name the component — explain the material choice.

| # | Component            | Function                              | Material(s)                             | Why is this material suitable?                                |
|---|----------------------|---------------------------------------|-----------------------------------------|---------------------------------------------------------------|
| 1 | LCD display          | Display video and UI                  | Glass, polymers, semiconductors         | It provide a clear display surface to enable display control. |
| 2 | DVD Optical Drive    | Reads data from DVD/CD using laser    | Plastics, metals, electronic components | Lightweight and durability                                    |
| 3 | Optical Pickup/Laser | Reads information stored on the disc  | Semiconductor + plastic material        | Highly accurate reading of data from the disc                 |
| 4 | PCB                  | Controls processing & power           | Copper, semiconductors                  | Good electronic processing and control                        |
| 5 | Speakers             | Converts electrical signal into sound | Plastic, rubber                         | lightweight, insulation and flexibility                       |
| 6 | Plastic Housing      | Internal components                   | Engineering Plastic                     | Durable, electrically insulation                              |

Engineering Challenge: Which component demonstrates the most interesting material selection? Why?

LCD display because it combines glass, polymers and semiconductor materials to produce a clear image while keeping the device lightweight and durable

## Part C — Material Detective

Tick the major material categories present in your device:

- ☒ Ferrous metals   ☒ Aluminium   ☒ Copper   ☐ Precious metals  
☒ Engineering plastics   ☒ Glass   ☒ Ceramics   ☒ Semiconductor materials  
☒ Rubber/elastomers   ☐ Other: \_\_\_\_\_

|                   | Material 1                                             | Material 2                                     |
|-------------------|--------------------------------------------------------|------------------------------------------------|
| Name              | Copper                                                 | Engineering Plastic                            |
| Where is it used? | Power cables, PCB tracks, connectors & internal wiring | Outer casing, buttons, disc, internal supports |
| Why is it used?   | Because copper conducts electricity                    | Its lightweight, strong and inexpensive        |

|                                    | Material 1                                            | Material 2                                     |
|------------------------------------|-------------------------------------------------------|------------------------------------------------|
| Can it be recovered/recycled? How? | Yes, Copper parts can be separated during dismantling | Yes, after separating different plastic types. |

Part D — Hidden Value: The Urban Mine

Identify the three most valuable/recoverable materials in your device.

| Material           | Component containing it                        | Potential value            | Recovery difficulty        |
|--------------------|------------------------------------------------|----------------------------|----------------------------|
| 1. Copper          | PCB, wires, speakers & connectors              | Low / Medium / <u>High</u> | Low / <u>Medium</u> / High |
| 2. Aluminium       | Heat dissipation parts & electronic components | <u>Low</u> / Medium / High | Low / Medium / <u>High</u> |
| 3. Precious metals | PCB & electronic contacts                      | Low / Medium / <u>High</u> | Low / Medium / <u>High</u> |

Critical Thinking: Is the most valuable material necessarily the easiest to recover? Explain.

No. Precious metals may have high value but are present in very small quantities and are embedded in electronic components.

Part E — Hazard Detective

Identify components that may require controlled handling.

| Component            | Possible hazard                       | Risk if improperly disposed  | Handling requirement  |
|----------------------|---------------------------------------|------------------------------|-----------------------|
| Rechargeable battery | Can leak, short circuit or be damaged | Fire risk                    | Controlled recycling  |
| PCB                  | Contains various electronic materials | Pollution & resource loss    | E-waste recycling     |
| LCD screen           | Contains glass & electronic layers    | Can break & create difficult | Handle carefully      |
| Plastic casing       | Improper burning & harmful fumes      | Air pollution                | Recycle, do not burn. |

Most environmentally concerning component: \_\_\_\_\_

Why? Rechargeable battery.  
The battery contains electrochemical materials and must be handled separately. Improper disposal can cause leakage, fire hazards and environmental contamination.

Part F — End-of-Life Decision

Recommend the most appropriate pathway for each major component.

| Component    | Reuse                    | Repair                   | Refurbish                           | Recycle                             | Controlled Disposal      |
|--------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
| LED Display  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| DVD Drive    | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| PCB          | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| Copper wires | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

Justify ONE of your decisions: Battery → Controlled Disposal: Recycling.  
 The battery should not be thrown into normal household waste because it contains chemical and metallic materials.

## Part G — Design for Disassembly

Imagine that you are the engineer who designed this product. Identify three design features that make recycling, repair or disassembly difficult.

| #         | Existing design problem                                                  |
|-----------|--------------------------------------------------------------------------|
| Problem 1 | The outer casing is difficult to open for accessing internal components. |
| Problem 2 | The battery may be difficult to remove or replace.                       |
| Problem 3 | Different materials and electronic modules are tightly packed together.  |

Now redesign it!

Propose three engineering modifications that would make the product more circular.

| Existing Design Problem | Proposed Design Change                               | Sustainability Benefit                        |
|-------------------------|------------------------------------------------------|-----------------------------------------------|
| The Outer Design        | Standard screws instead of permanent clips           | Makes repair and dismantling easier           |
| Battery                 | Replaceable battery module with accessible connector | Failed battery would not make device unusable |
| Electric modules        | Modular internal design                              | Component replacement & material recovery     |

Sketch your redesigned device/component — show at least one change that improves repair, disassembly or material recovery.

| ORIGINAL | REDESIGNED |
|----------|------------|
|----------|------------|

Device Anatomy 2.0 — E-Waste & Environmental Management — TE INFT

|                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                               |                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <div style="border: 1px solid black; padding: 2px; margin-bottom: 2px;">LCD Screen</div> <div style="border: 1px solid black; padding: 2px; margin-bottom: 2px;">DVD Drive</div> <div style="border: 1px solid black; padding: 2px;">PCB + Battery</div> | <div style="border: 1px solid black; padding: 2px; margin-bottom: 2px;">LCD Screen</div> <div style="border: 1px solid black; padding: 2px; margin-bottom: 2px;">DVD Drive</div> <div style="border: 1px solid black; padding: 2px;">Battery Module<br/>(Removable PCB)</div> | Adj:<br>Screw-open casing<br>Replaceable battery<br>Modular component<br>Easy recycling |
| Problem:- Glued casing<br>Fixed Battery<br>Difficult repair                                                                                                                                                                                              |                                                                                                                                                                                                                                                                               |                                                                                         |

## Part H — E-Waste Journey

Trace the end-of-life journey of ONE component.

Component selected: Rechargeable Battery

| DEVICE | CONSUMER DISCARD | COLLECTION | DISMANTLING | MATERIAL RECOVERY | NEW PRODUCT / RESOURCE |
|--------|------------------|------------|-------------|-------------------|------------------------|
|--------|------------------|------------|-------------|-------------------|------------------------|

What could go wrong at any stage?

If the battery is thrown into regular waste, it may be damaged, leak or create a fire hazard, while valuable materials are also lost.

## Part I — Circular Economy Decision

Our device should NOT simply be treated as waste because... It contains valuable materials and components such as copper, aluminium, plastics, electronic components etc. They can potentially be repaired and, recovered or reused.

Which circular strategy would have the greatest impact?

☐ Repair ☐ Reuse ☒ Refurbishment ☐ Remanufacturing

☐ Recycling ☐ Material recovery

Why?

A portable DVD player can potentially be refurbished by replacing the battery, optical drive or damaged components while keeping the main body in use. This can significantly extend its useful life and reduce e-waste.

## Part J — Engineering Viva

Be prepared to defend your answers. Your instructor may ask any of the following:

1. Why was a particular material selected for one component? → Cu
2. Which material has the greatest recovery potential? → Cu
3. Which component is most difficult to recycle and why? → PCB
4. What is the single most effective design modification you propose?
5. If your sustainable redesign increases manufacturing cost by 8%, would you still recommend it? Why?

## Final Group Verdict

|                                                                                                                                                   |                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| The most valuable component is:                                                                                                                   | PCB and copper containing component                       |
| The component with the greatest environmental concern is:                                                                                         | Rechargeable battery                                      |
| The biggest design-for-recycling problem is:                                                                                                      | Difficult access to and separation of internal components |
| Our most important redesign recommendation is:                                                                                                    | Use a screw-open modular design with replaceable battery. |
| <b>ONE-SENTENCE ENGINEERING VERDICT</b><br>We recommend <u>redesigning the portable DVD player</u> because <u>it's useful to reduce e-waste</u> . |                                                           |

## Assessment Rubric — 10 Marks

| Criterion                       | Excellent (2)                                     | Good (1.5)             | Developing (1)        | Marks     |
|---------------------------------|---------------------------------------------------|------------------------|-----------------------|-----------|
| 1. Engineering Anatomy          | Accurate component + function + material analysis | Mostly accurate        | Basic identification  | /2        |
| 2. Material & Recovery Analysis | Strong material/recovery reasoning                | Adequate reasoning     | Limited reasoning     | /2        |
| 3. Hazard & E-waste Assessment  | Identifies risks with clear justification         | Identifies major risks | Basic identification  | /2        |
| 4. Design for Disassembly       | Innovative, feasible engineering solutions        | Relevant solutions     | Basic suggestions     | /2        |
| 5. Engineering Defence/Viva     | Clear, evidence-based justification               | Reasonable explanation | Limited justification | /2        |
| <b>TOTAL</b>                    |                                                   |                        |                       | <b>10</b> |

## Reflection

What surprised you most about the materials hidden inside your device?

I was surprised that small portable DVD player contains many different materials and components.

After this activity, would you still call this device "waste"? Why?

No, because many of its components and materials can potentially be reused, repaired or recovered.

### KEY TAKEAWAY

E-waste is not merely waste — it is a complex mixture of materials, components, resources and risks that requires engineering solutions for responsible recovery.

